AIR STANDARD CYCLES

CONTENTS

- Introduction
- Air Standard Cycles
- Otto cycle (Thermodynamic Analysis of Air-Standard Otto Cycle)
- Diesel Cycle (Thermodynamic Analysis of Air-Standard Diesel Cycle)
- Dual Cycle (Thermodynamic Analysis of Air-Standard Dual Cycle)
- Comparison of Otto, Diesel, and Dual Cycles

DEFINITION:

- ***An air standard cycle*** is an idealized thermodynamic cycle used to analyse the performance of internal combustion engines. It simplifies the actual engine cycle by making certain assumptions

ASSUMPTIONS OF AIR STANDARD CYCLES

The following are the main assumptions of an air standard cycle:

- ***Air as the working fluid***: Air is assumed to be the working fluid throughout the cycle, and it circulates in a closed cycle.
- ***Ideal gas behaviour***: Air is assumed to behave as an ideal gas and obey the gas laws.
- ***Constant mass***: The mass of air remains constant throughout the cycle.
- ***No chemical reaction***: The combustion process is replaced by heat addition from an external source. Therefore, no chemical reaction takes place.

- ***Heat rejection***: The exhaust process is replaced by heat rejection to an external sink, which brings the working fluid back to its initial state.
- ***Reversible processes***: All processes in the cycle are assumed to be internally reversible.
- ***No friction***: Friction losses and other mechanical losses are neglected.
- ***Specific heat assumption***: The specific heat of air is assumed to remain constant during the cycle. This is known as the cold air standard assumption.

Additional assumptions under Cold Air Standard Cycle

- Air is assumed to have constant specific heats.
- The specific heat values are taken at room temperature, usually 25°C (298 K).
- The ratio of specific heats is constant.

Air Standard Cycles

Air-Standard cycle differs from the actual by the following:

- The working fluid throughout the cycle is *air*, and this is treated as a *perfect gas.*
- The *compression* process 1–2 and the *expansion* process 3–4 are both treated as *frictionless and adiabatic* (without heat loss).
- In place of *heat addition by internal combustion*, phase 2–3 of the process is represented by *the addition of heat from an external source* and not from chemical reactions during the cycle.
- All the processes in (ideal) power cycles are *internally reversible*.

5. Some heat is assumed to be rejected to a constant low temperature sink during the cycle.
6. It is assumed that there are *no heat losses* from the system to the surroundings.
7. The working medium has *constant specific heats* throughout the cycle.
8. The physical constant viz., *C_p, C_v, γ and M of working medium are the same as those of air standard atmospheric conditions*. For example

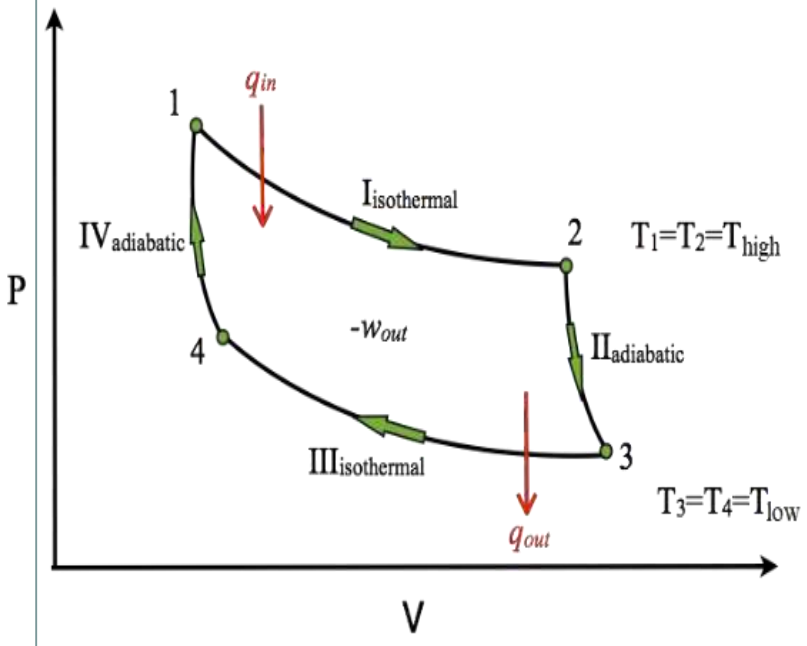
C_p	=	1.005 kJ/kg K
C_v	=	0.717 kJ/kg K
γ	=	1.4
M	=	29 kg/kmole

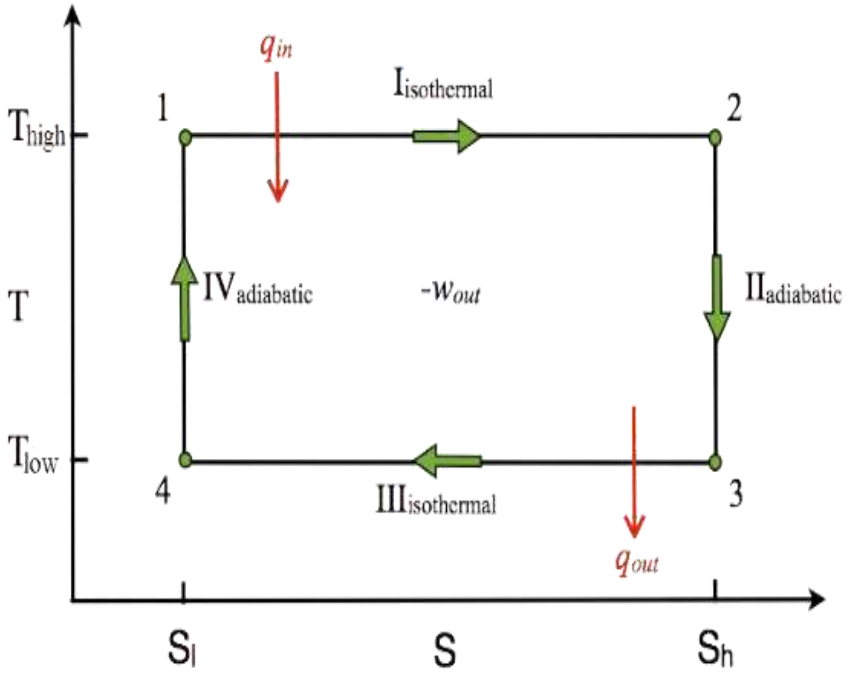
Air Standard Cycles

- Carnot Cycle
- Otto Cycle
- Diesel Cycle
- Dual Cycle

Carnot cycle

- This cycle consists of **two isothermal and reversible Adiabatic processes**.
- Carnot cycle is a perfect cycle and engines can be compared with it to judge the degree of perfection.





1-2:reversible isothermal expansion

2-3:reversible adiabatic expansion (isentropic expansion)

3-4:reversible isothermal compression

4-1:reversible adiabatic compression (isentropic compression)

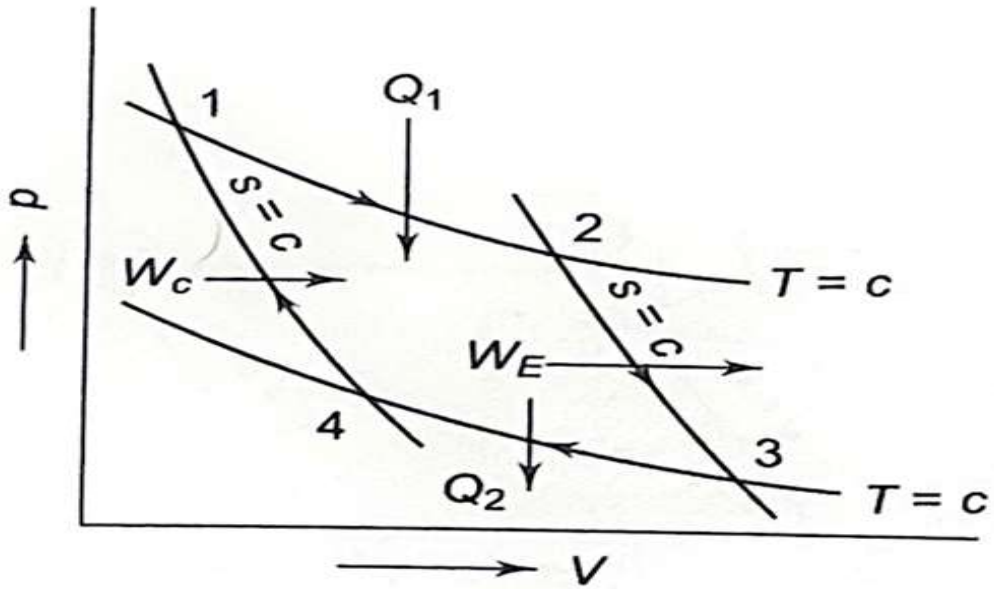
DERIVATION

$$Q_{1-2} = RT_1 \ln \frac{v_2}{v_1}; \quad W_{1-2} = RT_1 \ln \frac{v_2}{v_1}$$

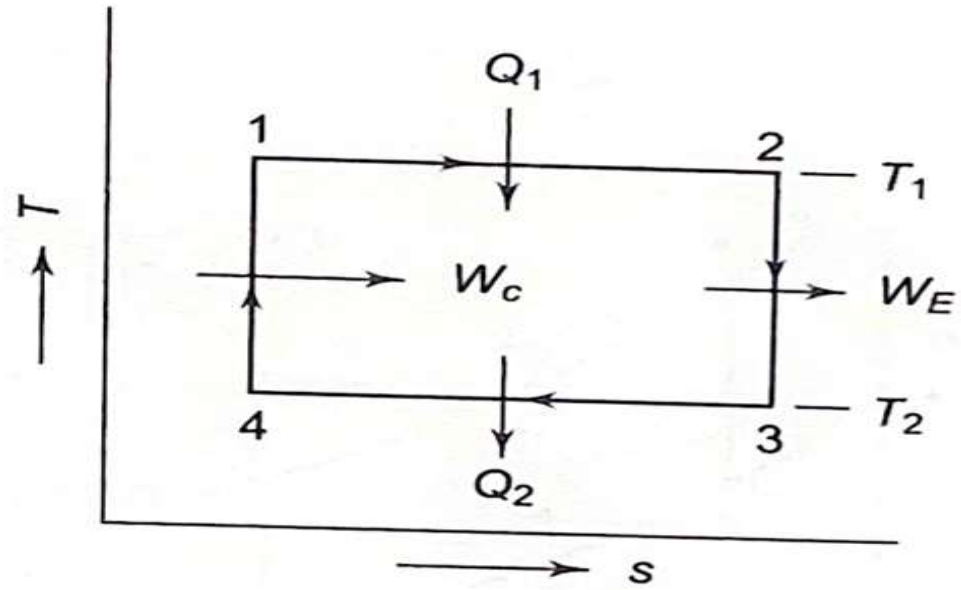
$$Q_{2-3} = 0; \quad W_{2-3} = -c_v (T_3 - T_2)$$

$$Q_{3-4} = RT_2 \ln \frac{v_4}{v_3}; \quad W_{3-4} = RT_2 \ln \frac{v_4}{v_3}$$

$$Q_{4-1} = 0; \quad W_{4-1} = -c_v (T_1 - T_4)$$



(a)



(b)

$$\frac{v_2}{v_3} = \left(\frac{T_2}{T_1}\right)^{1/(\gamma-1)}$$

$$\frac{v_1}{v_4} = \left(\frac{T_2}{T_1}\right)^{1/(\gamma-1)}$$

$$\frac{v_2}{v_3} = \frac{v_1}{v_4} \text{ or } \frac{v_2}{v_1} = \frac{v_3}{v_4}$$

$$Q_1 = \text{Heat added} = RT_1 \ln \frac{v_2}{v_1}$$

$$W_{net} = Q_1 - Q_2 = R \ln \frac{v_2}{v_1}(T_1 - T_2)$$

$$\eta_{cycle} = \frac{W_{net}}{Q_1} = \frac{T_1 - T_2}{T_1} = 1 - \frac{T_2}{T_1} = 1 - \frac{T_{lower}}{T_{higher}}$$

1. **Isochoric Process:** For this process, $V = C$

   $\therefore \quad P \propto T$

   $\Rightarrow \quad \frac{P_1}{T_1} = \frac{P_2}{T_2}$

   This is also known as Gay-Lussac's law for ideal gases.

2. **Isobaric Process:** For this process, $P = C$

   $\therefore \quad V \propto T$

   $\Rightarrow \quad \frac{V_1}{T_1} = \frac{V_2}{T_2}$ or $\frac{v_1}{T_1} = \frac{v_2}{T_2}$ [v_1 and v_2 are specific volumes]

   This is also known as Charle's law for ideal gases.

3. **Isothermal Process:** For this process $T = C$

   $P \propto \frac{1}{V}$

   $\Rightarrow \quad P_1V_1 = P_2V_2$ or $P_1v_1 = P_2v_2$

   This is also known as Boyle's law for ideal gases.

4. **Adiabatic Process:** For this process, $PV^\gamma = C$

   The relation between two states comes out to be

   $$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} \quad \text{or} = \left(\frac{v_1}{v_2}\right)^{\gamma-1}$$

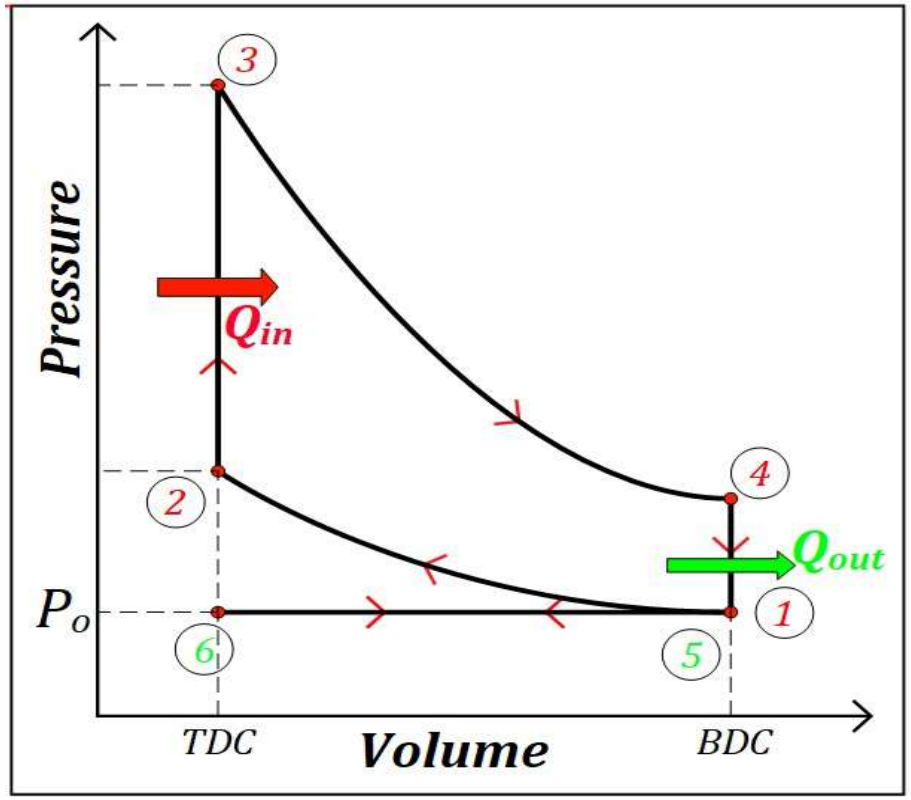
5. **Polytropic Process:** For this process, $PV^n = C$

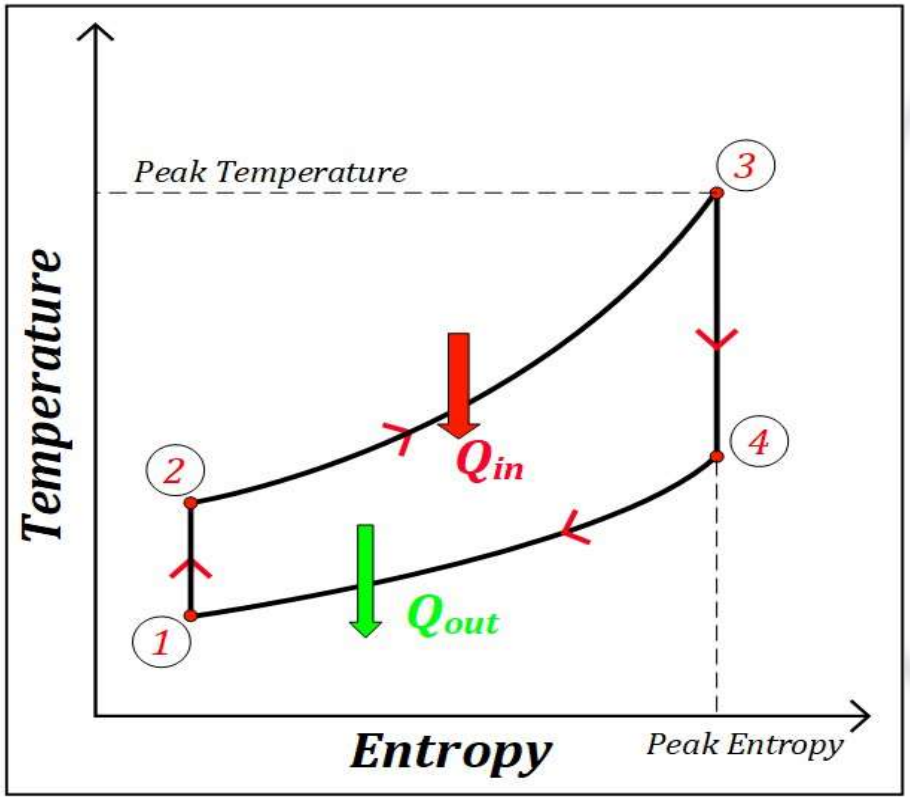
   Now, for ideal gas, $P^{1-n}T^n = C$ and $TV^{n-1} = C$

   Similarly,
   $$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} = \left(\frac{V_1}{V_2}\right)^{n-1} \quad \text{or} = \left(\frac{v_1}{v_2}\right)^{n-1}$$

Air Standard Otto Cycle

- The air-standard-Otto cycle is the idealized cycle for the spark-ignition internal combustion engines. The Otto cycle



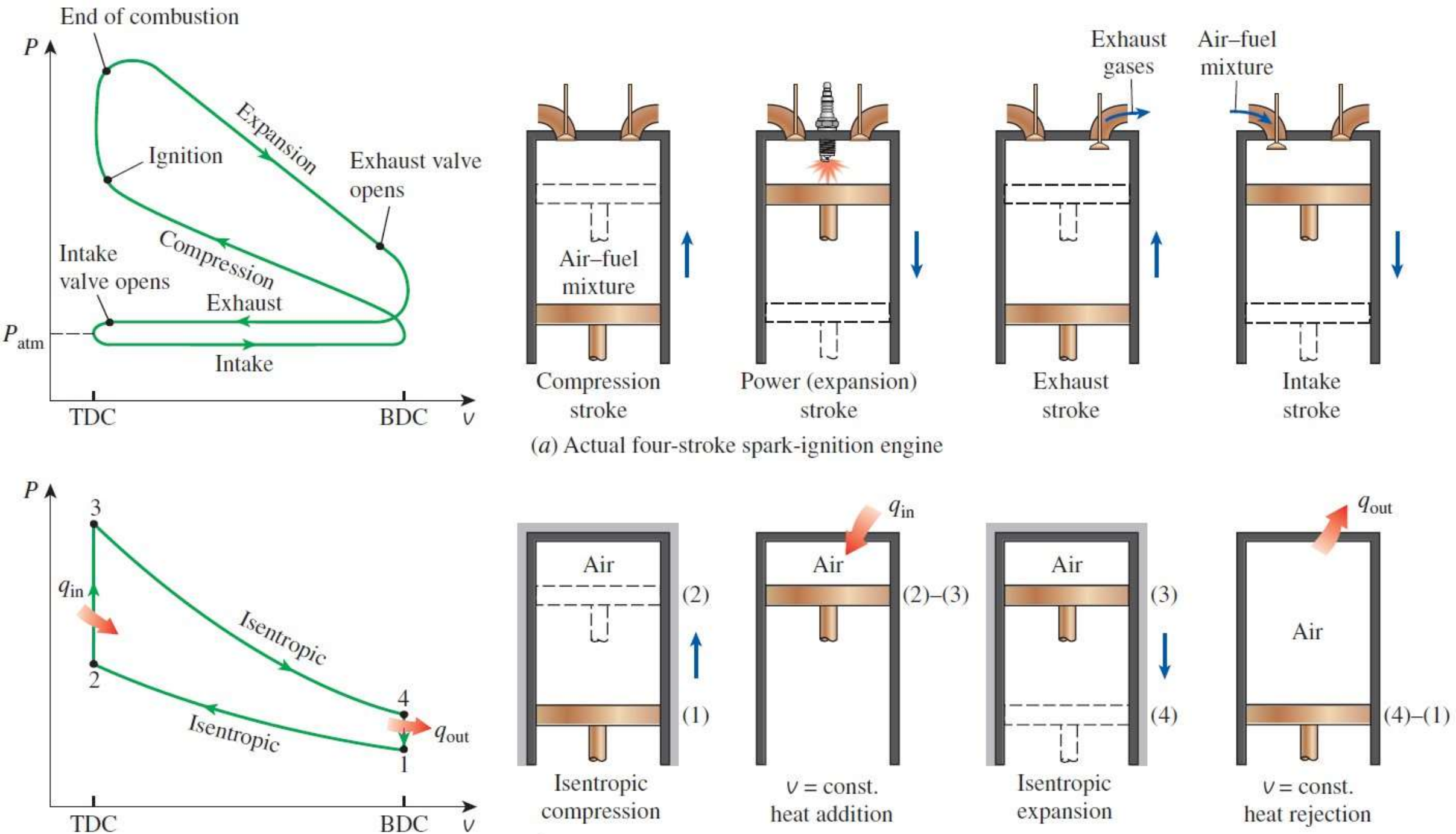


1-2: Reversible adiabatic compression (isentropic compression)

2-3: Heat addition at constant volume.

3-4: Reversible adiabatic expansion(isentropic expansion)

4-1: Heat rejection at constant volume.


End of combustion
P
Expansion
Ignition
Exhaust valve opens
Compression
Intake valve opens
Exhaust
P_{atm}
Intake
TDC
BDC
v
Air–fuel mixture
Compression stroke
Power (expansion) stroke
Exhaust gases
Air–fuel mixture
Exhaust stroke
Intake stroke
(a) Actual four-stroke spark-ignition engine
P
3
q_{in}
Isentropic
2
4
q_{out}
Isentropic
1
TDC
BDC
v
Air
(2)
(1)
Isentropic compression
q_{in}
Air
(2)–(3)
v = const. heat addition
Air
(3)
(4)
Isentropic expansion
q_{out}
Air
(4)–(1)
v = const. heat rejection
(b) Ideal Otto cycle

Otto Cycle

The **intake stroke** of the Otto cycle starts with the piston at TDC and is a constant-pressure process at an inlet pressure of one atmosphere **(6-1)**.

For real engine, inlet pressure will be at a pressure slightly less than atmospheric due to pressure losses in the inlet air flow. The temperature of the air during the inlet stroke is increased as the air passes through the hot intake manifold.

The **second stroke** of the cycle is an **Isentropic compression** from BDC to TDC **(1-2)**.

In a real engine, the beginning of the stroke is affected by the intake valve not being fully closed until slightly after BDC. The end of compression is affected by the firing of the spark plug before TDC.

There is an increase in pressure during the compression stroke, but the temperature within the cylinder is increased substantially due to compressive heating.

The **third stroke** of the cycle is a **Constant-Volume heat input** process **2-3** at TDC.

In real engine, constant volume process is replaced by Combustion process which occurs at nearly Constant volume conditions. In a real engine combustion is started slightly before TDC, reaches its maximum speed near TDC, and is terminated a little after TDC.

During combustion, a large amount of heat is added which results in peak temperature and peak pressure of air at the end of combustion.

Otto Cycle

In the **fourth stroke**, high Pressure and Enthalpy produces **power stroke** which pushes the piston backwards (**3-4**).
In a real engine, it is approximated as an **Isentropic expansion**, the beginning of the power stroke is affected by the last part of the combustion process. The end of the power stroke is affected by the exhaust valve being opened before BDC.
During the power stroke, values of both the temperature and pressure within the cylinder decrease as volume increases from TDC to BDC.

In the **fifth stroke**, near the end of power stroke, exhaust valve is opened and **exhaust blowdown** occurs.
Pressure is reduced to exhaust manifold and large amount of enthalpy is carried away by exhaust gases. The exhaust valve is opened before BDC to allow for the finite time of blowdown to occur.
In real cycle, exhaust blowdown is replaced with a **Constant-Volume heat rejection**, closed-system process **4-5**. Pressure at the end has been reduced to atmospheric pressure.

The **last stroke** of the four-stroke cycle now occurs as the piston travels from BDC to TDC.
Process **5-6** is the exhaust stroke that occurs at a constant pressure of one atmosphere due to the open exhaust valve.

$$\eta_{th} = \frac{\text{Net workdone}}{\text{Net heat added}}$$

Since processes 1-2 and 3-4 are adiabatic processes, the heat transfer during the cycle takes place only during processes 2-3 and 4-1 respectively. Therefore, thermal efficiency can be written as,

$$\eta_{th} = \frac{\text{Heat added - Heat rejected}}{\text{Heat added}}$$

Consider ‘m’ kg of working fluid,

$$\text{Heat added} = mC_V(T_3 - T_2)$$

$$\text{Heat Rejected} = mC_V(T_4 - T_1)$$

$$\eta_{th} = \frac{mC_V(T_3 - T_2) - mC_V(T_4 - T_1)}{mC_V(T_3 - T_2)} = 1 - \frac{T_4 - T_1}{T_3 - T_2}$$

- For the reversible adiabatic processes 3-4 and 1-2, we can write,

$$\frac{T_4}{T_3} = \left(\frac{v_3}{v_4}\right)^{\gamma - 1} \quad \text{and} \quad \frac{T_1}{T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma - 1}$$

$$v_2 = v_3 \text{ and } v_4 = v_1$$

$$\frac{T_4}{T_3} = \frac{T_1}{T_2} = \frac{T_4 - T_1}{T_3 - T_2} = \left(\frac{V_2}{V_1}\right)^{\gamma - 1}$$

$$\eta_{th} = 1 - \frac{T_1}{T_2} = 1 - \left(\frac{V_2}{V_1}\right)^{\gamma - 1}$$

The ratio $\frac{V_1}{V_2}$ is called as compression ratio, r.

$$\eta_{th} = 1 - \left(\frac{1}{r}\right)^{\gamma - 1}$$

Various Processes undergone by an ideal gas

We know for ideal gas $PV = mRT$

or $\frac{PV}{T} = mR =$ constant (as mass is always conserved)

$\therefore \frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$ (for closed system)

or $\frac{P_1v_1}{T_1} = \frac{P_2v_2}{T_2}$ (for open/flow systems, where v_1 and v_2 are specific volumes)

1. **Isochoric Process:** For this process, $V = C$

   $\therefore P \propto T$

   $\Rightarrow \frac{P_1}{T_1} = \frac{P_2}{T_2}$

   This is also known as Gay-Lussac's law for ideal gases.

2. **Isobaric Process:** For this process, $P = C$

   $\therefore V \propto T$

   $\Rightarrow \frac{V_1}{T_1} = \frac{V_2}{T_2}$ or $\frac{v_1}{T_1} = \frac{v_2}{T_2}$ [v_1 and v_2 are specific volumes]

   This is also known as Charle's law for ideal gases.

3. **Isothermal Process:** For this process $T = C$

   $P \propto \frac{1}{V}$

   $\Rightarrow P_1V_1 = P_2V_2$ or $P_1v_1 = P_2v_2$

   This is also known as Boyle's law for ideal gases.

4. **Adiabatic Process:** For this process, $PV^\gamma = C$

   The relation between two states comes out to be

   $$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} \quad \text{or} = \left(\frac{v_1}{v_2}\right)^{\gamma-1}$$

5. **Polytropic Process:** For this process, $PV^n = C$

   Now, for ideal gas, $P^{1-n}T^n = C$ and $TV^{n-1} = C$

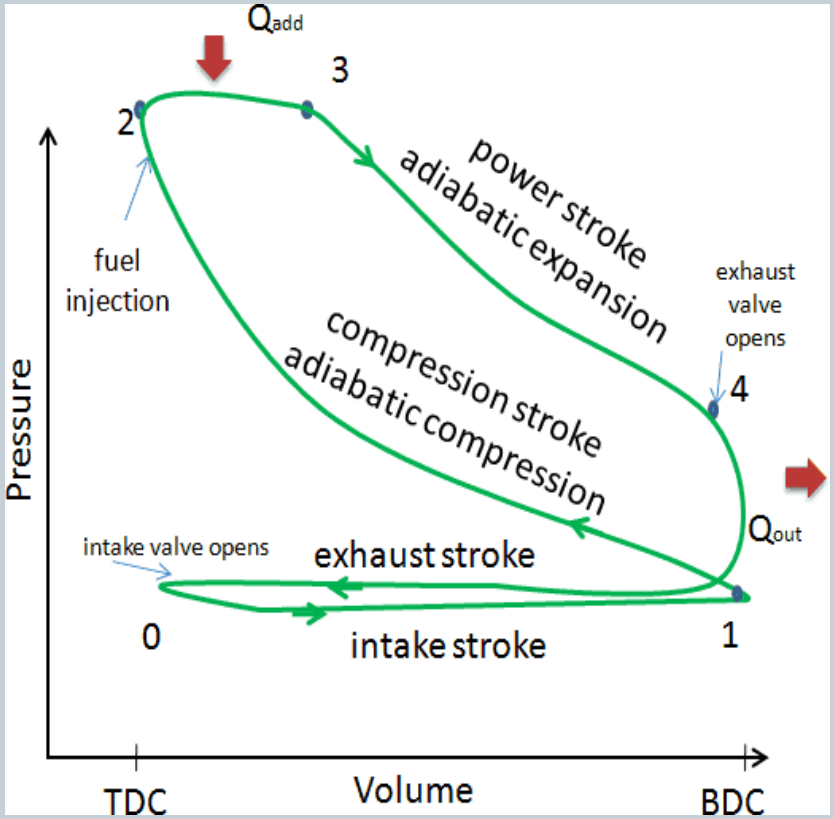
   Similarly,

   $$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} = \left(\frac{V_1}{V_2}\right)^{n-1} \quad \text{or} = \left(\frac{v_1}{v_2}\right)^{n-1}$$

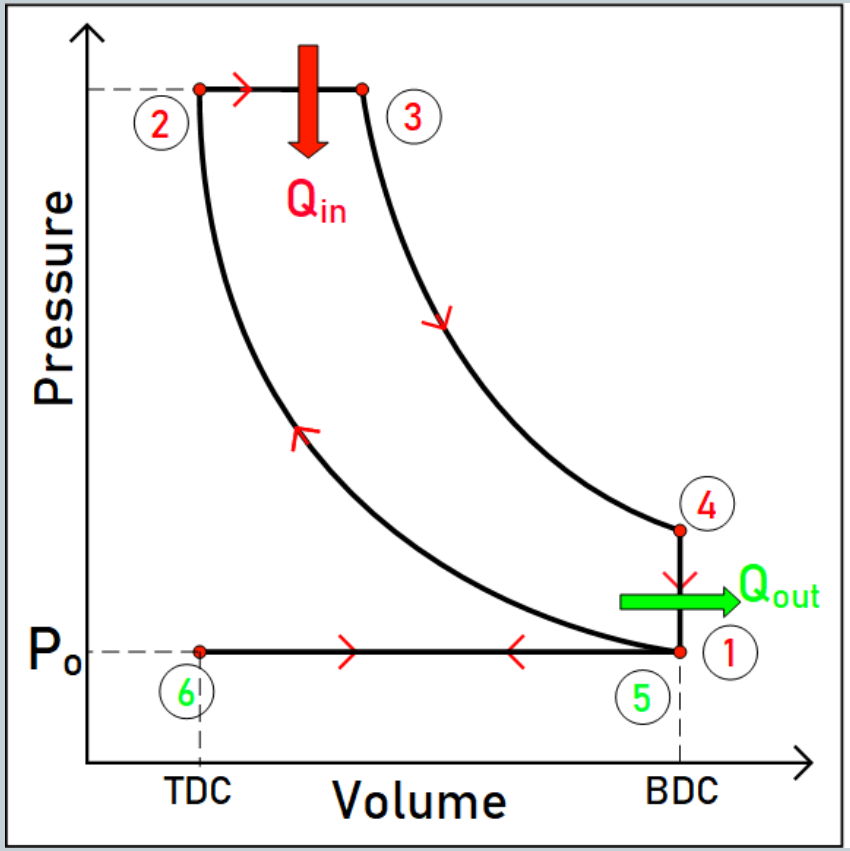
Diesel Cycle

- Early CI engines injected fuel into the combustion chamber very late in the compression stroke.
- Due to ignition delay and the finite time required to inject the fuel, combustion lasted into the expansion stroke. This kept the pressure at peak levels well past TDC.
- This combustion process is best approximated as a constant-pressure heat input in an air-standard cycle.
- In CI engines, the air is compressed to a temperature, that is above the auto ignition temperature of the fuel and combustion starts on the contact as the fuel is injected into the hot air.

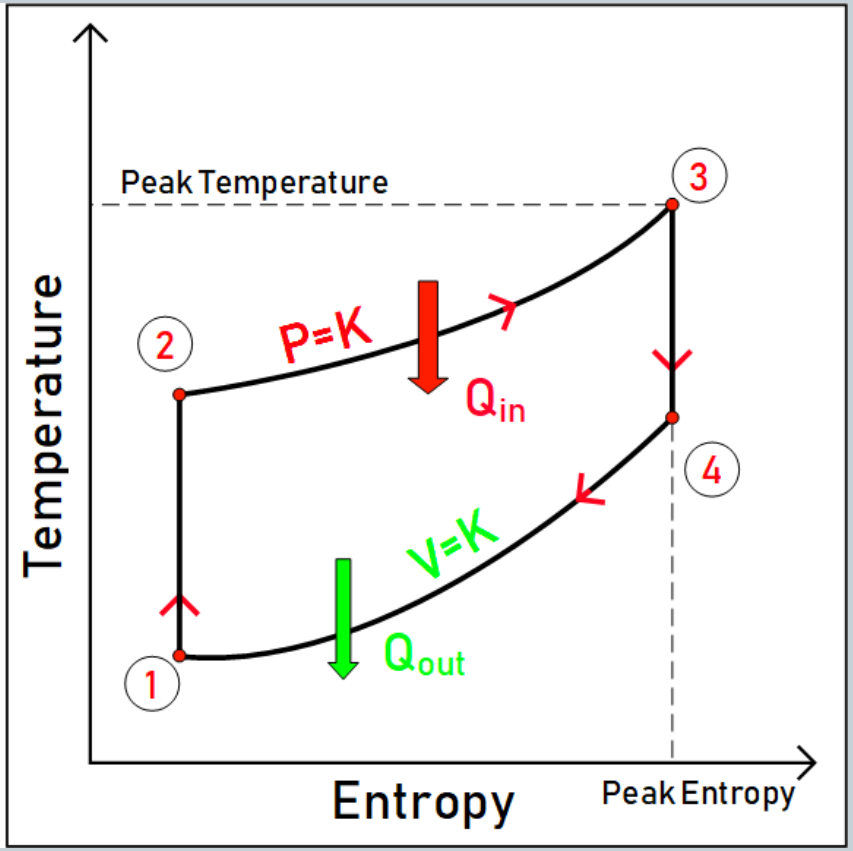
Diesel Cycle



Actual Diesel cycle



P-V Diagram



T-S Diagram

- Consider ‘m’ kg of working fluid. Since the compression and expansion processes are reversible adiabatic processes, we can write,

$$\text{Heat supplied} = m C_p (T_3 - T_2) = (h_3 - h_2)$$

$$\text{Heat rejected} = m C_v (T_4 - T_1) = (u_4 - u_1)$$

$$\text{Workdone} = m C_p (T_3 - T_2) - m C_v (T_4 - T_1)$$

$$\eta_{th} = \frac{m C_p (T_3 - T_2) - m C_v (T_4 - T_1)}{m C_p (T_3 - T_2)}$$

$$= 1 - \frac{1}{\gamma}\left(\frac{T_4 - T_1}{T_3 - T_2}\right)$$

$$T_2 = T_1 r^{\gamma-1} \; ; \; r = \frac{v_1}{v_2} = \frac{v_4}{v_2}$$

Where r is compression ratio

$$\frac{T_3}{T_2} = \frac{v_3}{v_2} = r_c = \text{cutoff ratio}$$

$$T_3 = r_c T_2 = r_c T_1 r^{\gamma-1}$$

$$T_4 = T_3\left(\frac{v_3}{v_4}\right)^{\gamma-1} = T_3\left(\frac{v_4}{v_3}\right)^{\gamma-1}$$

1. **Isochoric Process:** For this process, $V = C$

   $\therefore \quad P \propto T$

   $\Rightarrow \quad \frac{P_1}{T_1} = \frac{P_2}{T_2}$

   This is also known as Gay-Lussac's law for ideal gases.

2. **Isobaric Process:** For this process, $P = C$

   $\therefore \quad V \propto T$

   $\Rightarrow \quad \frac{V_1}{T_1} = \frac{V_2}{T_2}$ or $\frac{v_1}{T_1} = \frac{v_2}{T_2}$ [v_1 and v_2 are specific volumes]

   This is also known as Charle's law for ideal gases.

3. **Isothermal Process:** For this process $T = C$

   $P \propto \frac{1}{V}$

   $\Rightarrow \quad P_1V_1 = P_2V_2$ or $P_1v_1 = P_2v_2$

   This is also known as Boyle's law for ideal gases.

4. **Adiabatic Process:** For this process, $PV^{\gamma} = C$

   The relation between two states comes out to be

   $$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{\gamma-1}{\gamma}} = \left(\frac{V_1}{V_2}\right)^{\gamma-1} \quad \text{or} = \left(\frac{v_1}{v_2}\right)^{\gamma-1}$$

5. **Polytropic Process:** For this process, $PV^n = C$

   Now, for ideal gas, $P^{1-n} T^n = C$ and $TV^{n-1} = C$

   Similarly,
   $$\frac{T_2}{T_1} = \left(\frac{P_2}{P_1}\right)^{\frac{n-1}{n}} = \left(\frac{V_1}{V_2}\right)^{n-1} \quad \text{or} = \left(\frac{v_1}{v_2}\right)^{n-1}$$

$$= T_3\left(\frac{v_4}{v_2}.\frac{v_2}{v_3}\right)^{1-\gamma} = T_3\left(\frac{r}{r_c}\right)^{1-\gamma}$$

$$= r_c\ T_1\ r^{\gamma-1}\left(\frac{r}{r_c}\right)^{1-\gamma} \quad ; \quad T_4 = r_c^{\gamma}\ T_1$$

Hence, $$\eta_{th} = 1 - \frac{1}{\gamma}\left\{\frac{r_c^{\gamma}\ T_1 - T_1}{r_c\ r^{\gamma-1}\ T_1 - r^{\gamma-1}\ T_1}\right\}$$

$$= 1 - r^{1-\gamma}\left\{\frac{r_c^{\gamma} - 1}{\gamma(r_c - 1)}\right\}$$

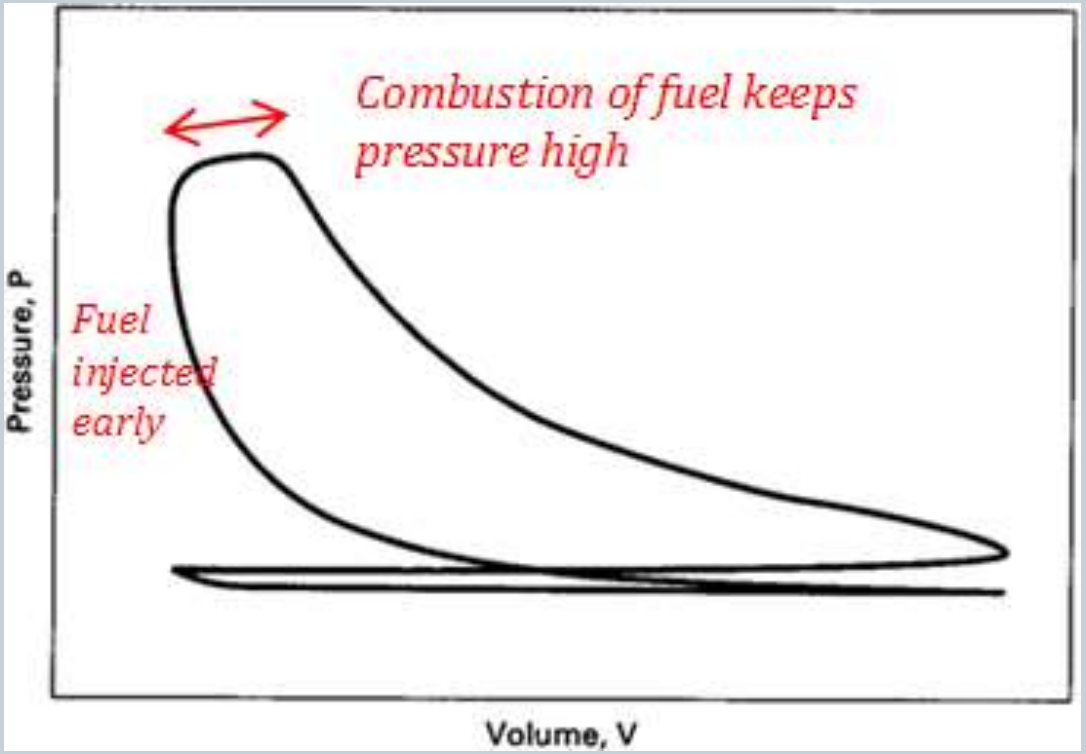
Where $r \rightarrow$ compression ratio, $r_c =$ cut off ratio

It is observed that, the thermal efficiency of the diesel engine can be increased by increasing the compression ratio, r, by decreasing the cut-off ratio, or by using a gas with large value of γ. Since the quantity $(r_\gamma\text{-}1)/\gamma(r_p\text{-}1)$ in above equation is always greater than unity, the efficiency of a Diesel cycle is always lower than that of an Otto cycle having the same compression ratio. However, practical Diesel engines uses higher compression ratios compared to petrol engines.

Difference between Actual Diesel and the Otto Engines

Otto Engine	Diesel Engine
1. Homogenous mixture of fuel and air formed in the carburetor is supplied to engine cylinder.	1. No carburetor is used. Air alone is supplied to the engine cylinder. Fuel is injected directly into the engine cylinder at the end of compression stroke by means of a fuel injector. Fuel-air mixture is heterogeneous.
2. Ignition is initiated by means of an electric spark plug.	2. No spark plug is used. Compression ratio is high and the high temperature of air ignites fuel.
3. Power output is controlled by varying the mass of fuel-air mixture by means of a throttle valve in the carburetor.	3. No throttle value is used. Power output is controlled only by means of the mass of fuel injected by the fuel injector.

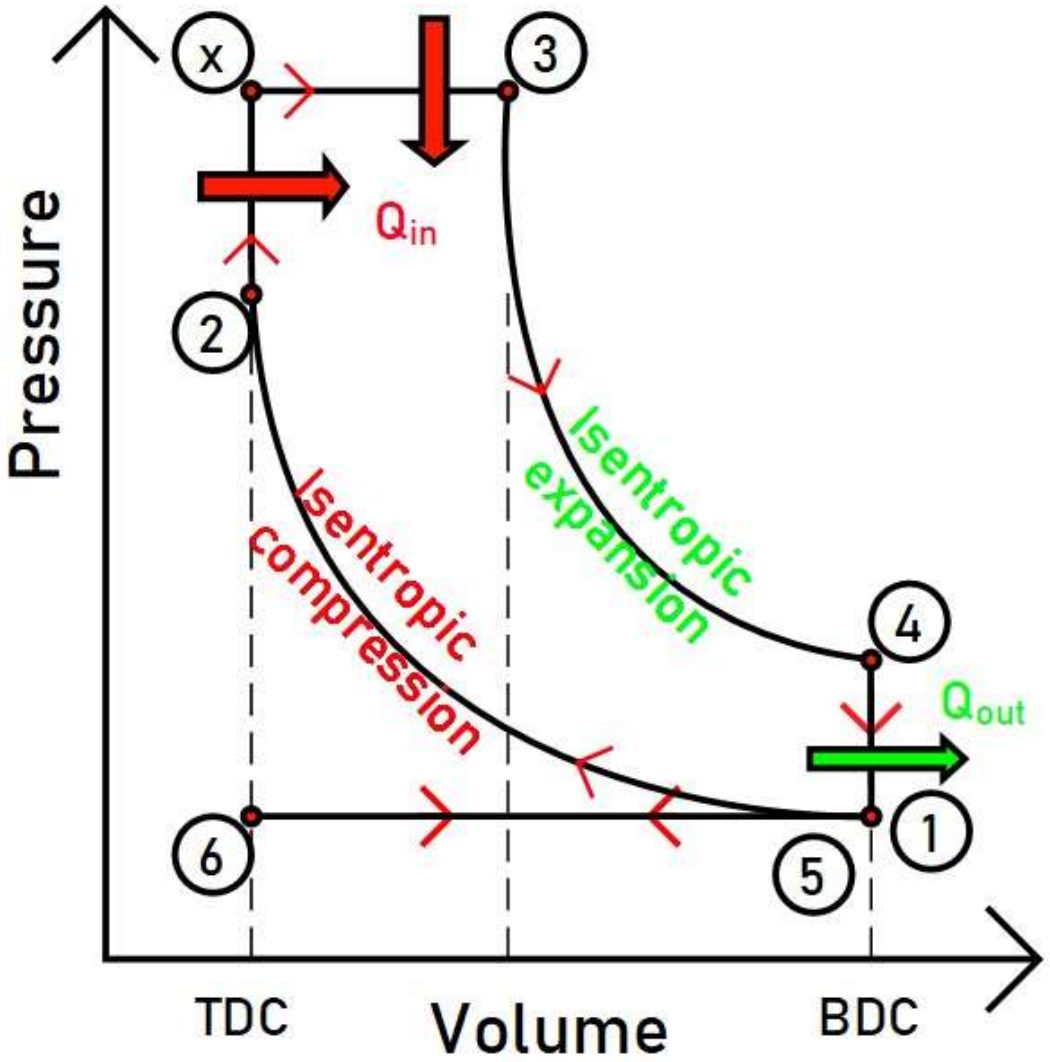
Dual Cycle

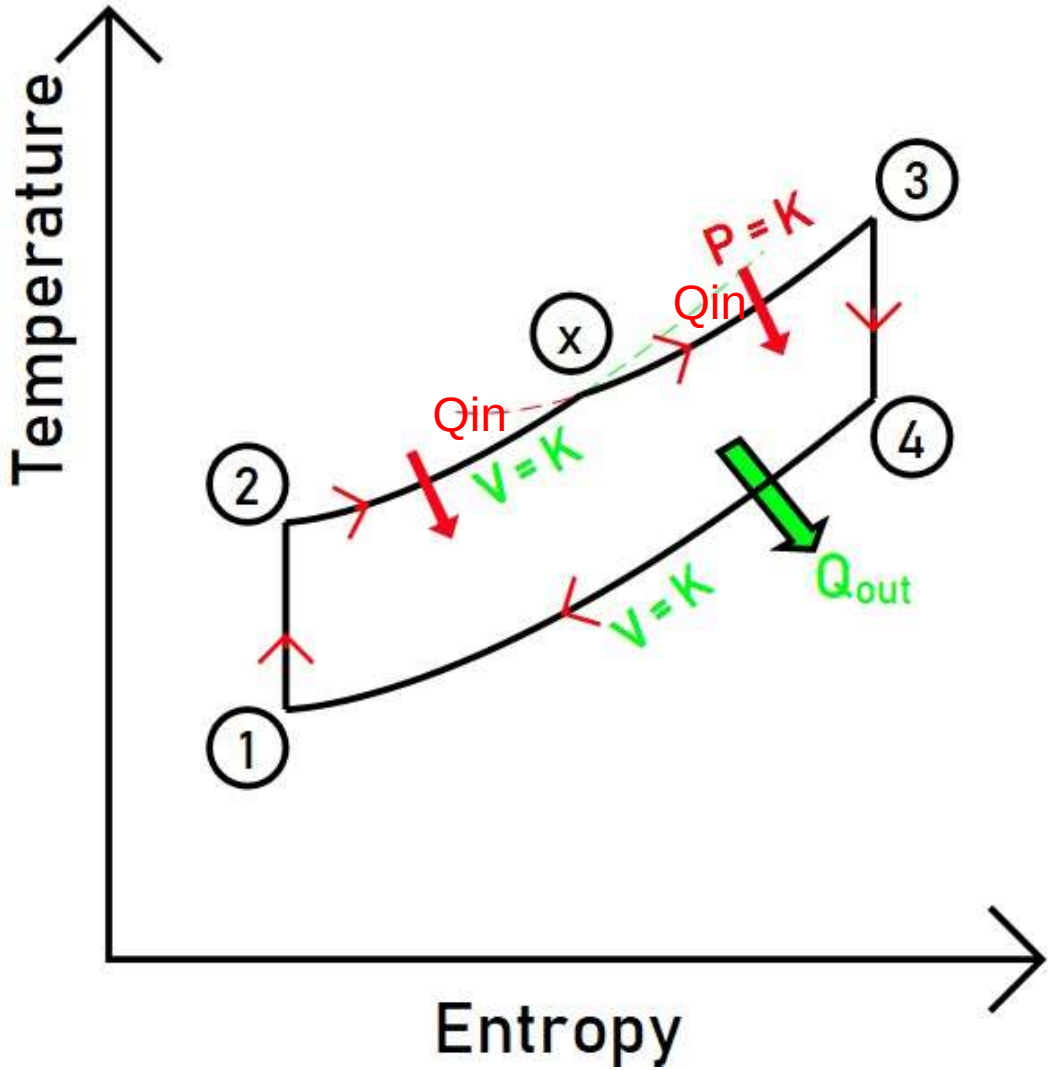


Indicator diagram for Dual Cycle

- The Dual cycle, also called a mixed cycle or limited pressure cycle, is a compromise between Otto and Diesel cycles.
- C.I engines have higher compression ratios and S.I engines have higher efficiencies.
- Instead of injecting fuel late in the compression stroke near TDC, modern C.I engines start to inject the fuel much earlier in the cycle somewhere around 20^0 TDC. So, part of combustion occurs at constant volume and remaining at constant pressure.
- Dual cycle is similar to Diesel cycle except for Heat input process.

Thermodynamic Analysis





Thermodynamic Analysis

Process 2-x

Constant volume Heat addition

$$w_{2-x} = 0$$

$$q_{2-x} = c_v \cdot (T_x - T_2) = u_x - u_2$$

$$\text{Pressure ratio } (r_p) = \frac{P_x}{P_2} = \frac{P_3}{P_2} = \frac{T_x}{T_2} = \frac{1}{r^k} \cdot \left(\frac{P_3}{P_1}\right)$$

Process x-3

Constant pressure heat addition

$$P_3 = P_x = P_{max}$$

$$q_{x-3} = c_p \cdot (T_3 - T_x) = h_3 - h_x$$

$$w_{x-3} = P_3 \cdot (v_3 - v_x)$$

$$T_3 = T_{max}$$

$$\text{Cut off ratio, } r_c = \frac{V_3}{V_x}$$

$$Q_{in} = Q_{2-x} + Q_{x-3} = m_f \cdot Q_{HV} \cdot \eta_c$$

$$q_{in} = q_{2-x} + q_{3-x} = (u_x - u_2) + (h_3 - h_x)$$

Thermodynamic Analysis

Thermal efficiency of dual cycle:

$$(\eta_t)_{Dual} = \frac{|w_{net}|}{|q_{in}|} = 1 - \left(\frac{|q_{out}|}{|q_{in}|}\right) = 1 - \frac{c_v(T_4 - T_1)}{\left[c_v(T_x - T_2) + c_p(T_3 - T_x)\right]}$$

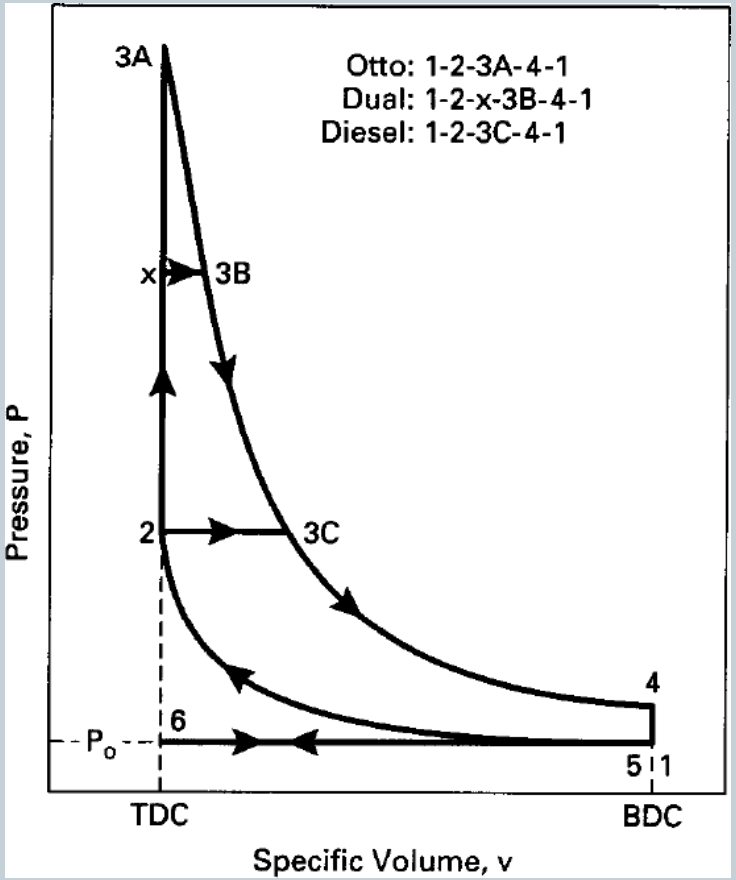
$$(\eta_t)_{Dual} = 1 - \frac{(T_4 - T_1)}{[(T_x - T_2) + \gamma \cdot (T_3 - T_x)]}$$

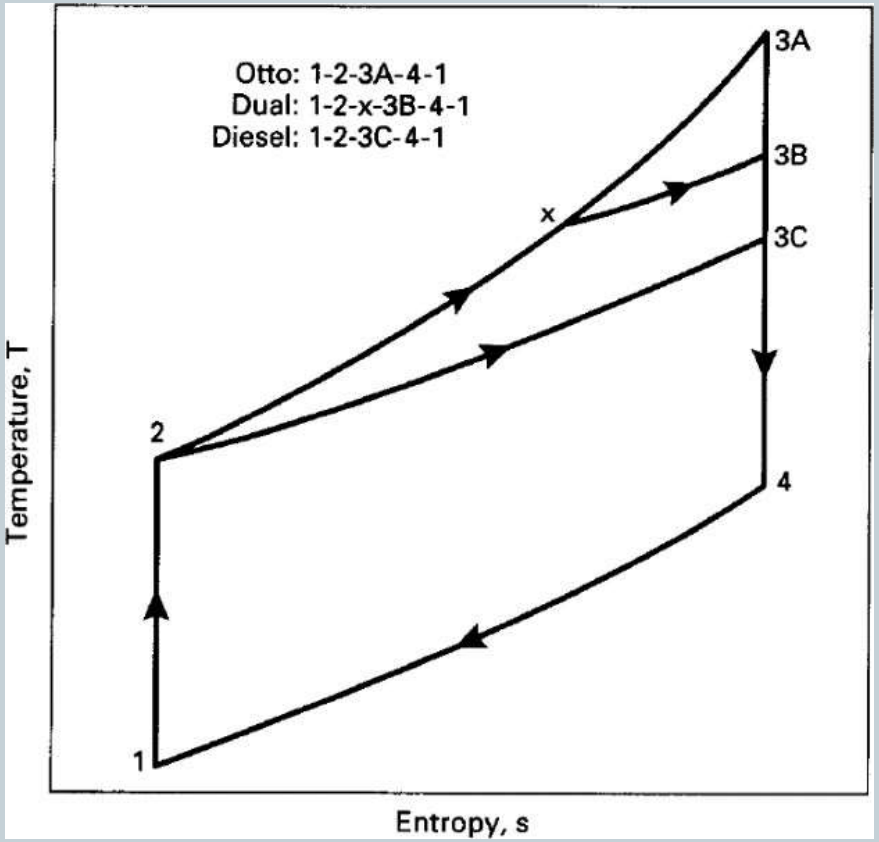
$$\boxed{(\eta_t)_{dual} = 1 - \frac{1}{r^{\gamma-1}} \cdot \left[\frac{r_p \cdot (r_c)^{\gamma} - 1}{\{\gamma \cdot r_p(r_c - 1) + r_p - 1\}}\right]}$$

$r_p \rightarrow$ pressure ratio, $r \rightarrow$ compression ratio, $r_c =$ cut off ratio

Comparison of Otto, Diesel and Dual Cycles

For the same Inlet conditions, the same compression ratios and same Heat removal.





$$(q_{in})_{otto} > (q_{in})_{dual} > (q_{in})_{diesel}$$

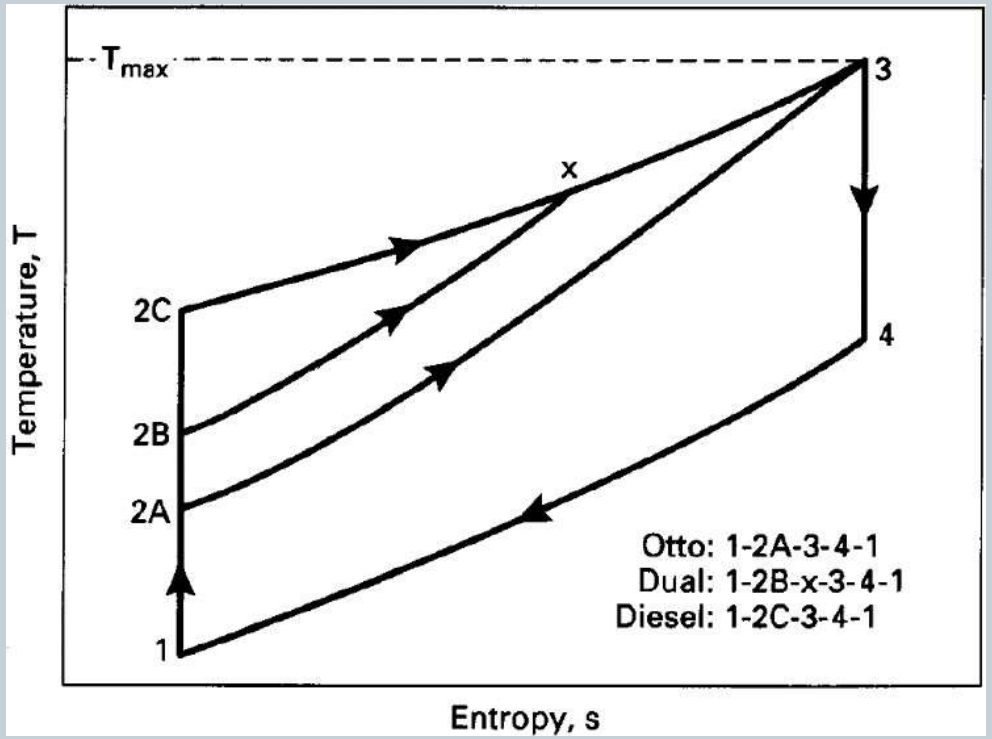
$$(\eta_t)_{otto} > (\eta_t)_{dual} > (\eta_t)_{diesel}$$

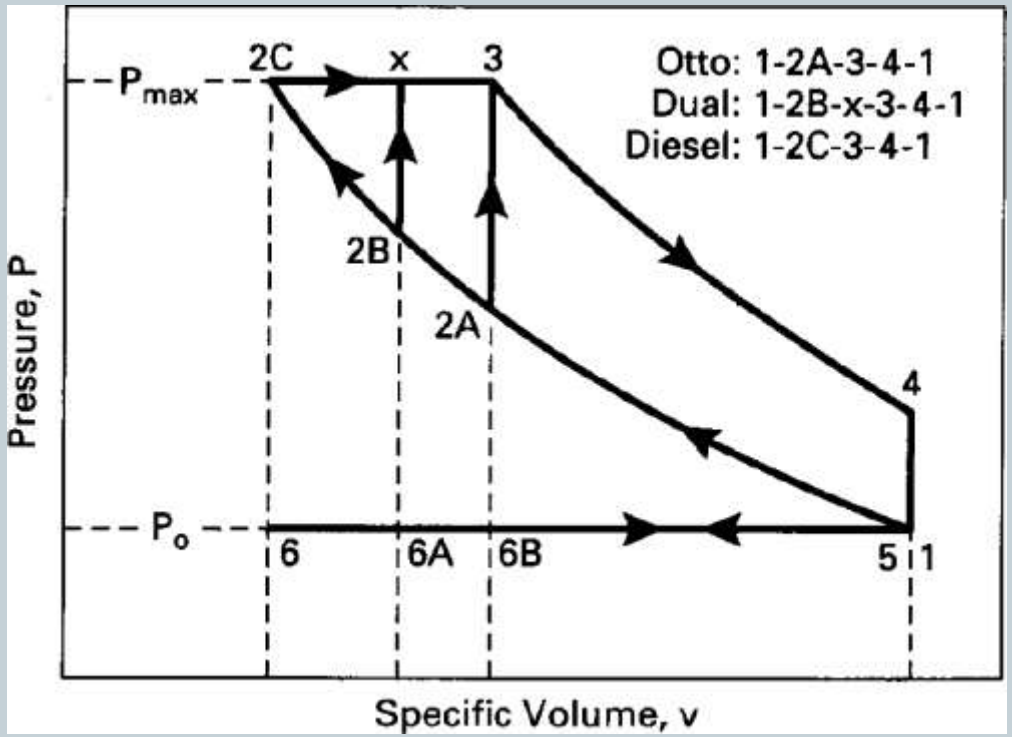
This is not a fair comparison because, as Compression ratio(r_c) of Diesel and Dual cycles are much higher than Otto cycle.

Area under the T-s graph is equal to Heat transfer. So, **q_{out} is same for 3 cycles.** But, **q_{in}** is different.

Comparison of Otto, Diesel and Dual Cycles

For the same inlet conditions, same peak pressure and same Heat removal.



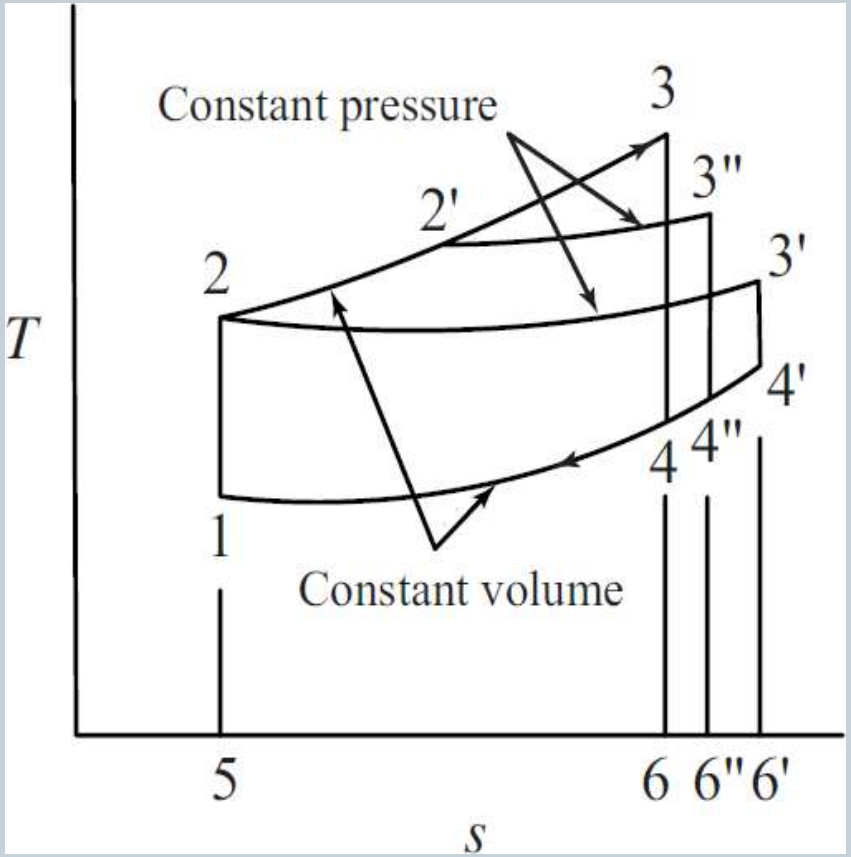


$$(q_{in})_{Diesel} > (q_{in})_{dual} > (q_{in})_{Otto}$$

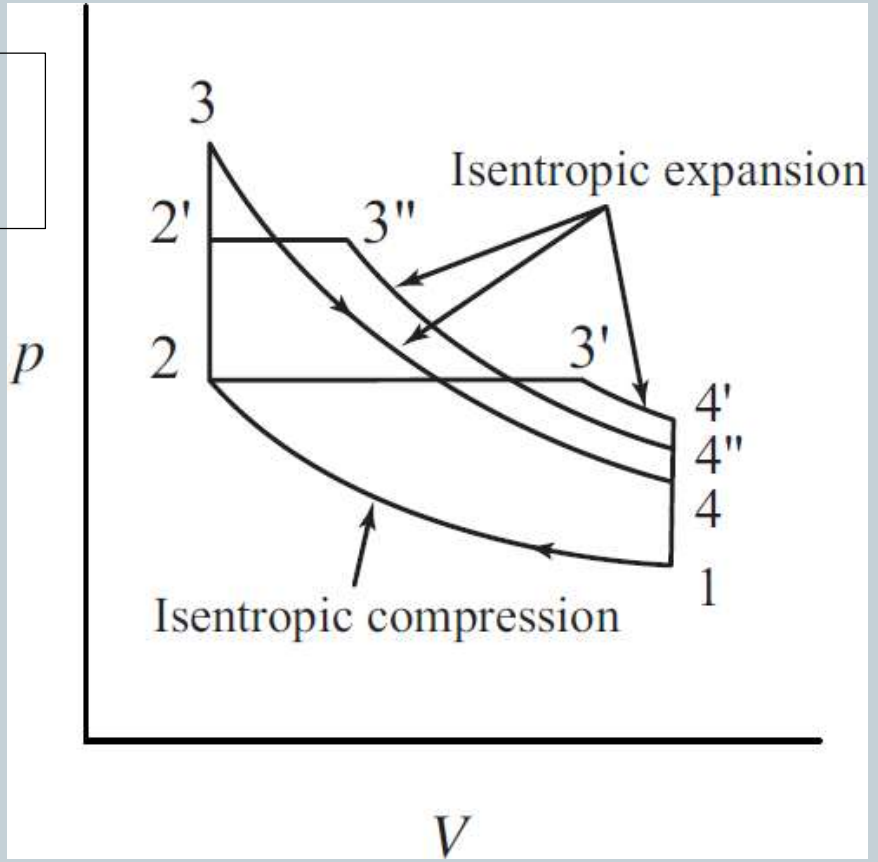
$$(\eta_t)_{Diesel} > (\eta_t)_{dual} > (\eta_t)_{Otto}$$

Comparison of Otto, Diesel and Dual Cycles

For the same Inlet conditions and Heat addition and compression ratio.



Otto cycle 1→2→3→4→1
Diesel cycle 1→2→3'→4'→1
Dual cycle 1→2→2'→3"→4"→1



$$(q_{out})_{otto} < (q_{in})_{dual} < (q_{in})_{diesel}$$

$$(\eta_t)_{otto} > (\eta_t)_{dual} > (\eta_t)_{diesel}$$